

Flood Digital Twin with Outcome-Grounded Tactical Critic

Product Overview | Hackathon Build Scope, dnhacks September 2026

Target User

- **Primary:** Emergency Operations Center (EOC) staff and incident-commander trainees who need to practice synthesizing hazard data into timely tactical decisions during a fast-moving flash flood.
- **Secondary:** Pre-deployment planners and strike-team leads who need a draft incident action plan (IAP) red-teamed against projected inundation and historical outcomes before a team leaves.
- **Tertiary:** Trainers and after-action reviewers who need an exportable, citation-backed decision log for debrief.

Problem

Kerr County's failure on 4 July 2025 was a decision-latency failure, not a data failure. A flood watch was issued at 1:18 p.m., a flash flood warning at 1:14 a.m., and a flash flood emergency at 4:03 a.m. Gauge data, radar-estimated rainfall, a county hazard mitigation plan rating a flood event likely within a year, and an emergency operations plan prescribing reconnaissance after a watch all existed.

- Nobody synthesized the available data into an evacuation order during the 2 hours 49 minutes between the warning and the emergency. Nearly 200 people died.
- Existing hazard tools (Hazus, NOAA National Water Model, NWS Flood Inundation Mapping) predict inundation but aren't integrated with tactical decision-making.
- Existing IAP/command software (Tablet Command, Adashi C&C;, WebEOC) records the tactics a human chose but doesn't know historical outcomes or propose alternatives.
- Existing exercise tools (Avalanche TTX, TabletopOne) log decisions but don't grade tactics against what happened in comparable real incidents.
- Nobody ships a usable tool that critiques tactics against historical outcomes — this is the specific, validated gap.

Proposed Solution

A flood-only digital twin that fuses physics-based inundation prediction with retrieval-grounded tactical critique, in two settings where a human commander still owns every operational decision:

- **Training mode** — a trainee runs the Kerr County replay timeline, makes tactical choices at fixed decision points (1:14 a.m., 2:30 a.m., 4:03 a.m.), and is critiqued against what comparable choices cost in the historical record, with every objection citing a source.
- **Pre-deployment red-team** — a draft IAP is checked against projected inundation and the AAR corpus before a strike team leaves.
- **Physics layer** — government hydrology models (HAND rasters, NOAA National Water Model streamflow) compute depth and velocity per zone; no primary reliance on a trained surrogate.

- **LLM/RAG layer** — a retrieval-grounded critic compares proposed tactics against an embedded AAR corpus (Kerr 2025, Wimberley 2015, Houston 2016, Llano 2018) and returns citation-backed objections; an output validator rejects any response citing a source that doesn't exist.
- **Visualization** — a synchronized 3D map (terrain + inundation + population/exposure overlays) driven by a single shared clock and session state, so the map, training-mode critique, and (later) chatbot can never contradict each other.

Explicitly positioned as a training and pre-deployment tool — not a live tactical recommender. A human commander makes every real operational call.

Build Order (MVP vs. Stretch vs. Post-MVP)

Ordered by what's required for a working, defensible demo. Everything above the line is core; everything below is additive and explicitly non-blocking.

Tier	Component	Why
MVP	Physics engine + impact extractor (hazard/exposure fusion)	Real, public hydrology data — the deterministic core everything else depends on.
MVP	AAR pipeline + RAG corpus + LLM critic + output validator	The core differentiator: citation-grounded tactical critique, not just a recorded decision.
MVP	3D map + training mode, synchronized via clock + session state	The demoable surface — shows physics and critique fused in one timeline-scrubbable view.
MVP	Search-area tool (probability-weighted polygons)	Completes the Kerr County replay's final decision point (4:03 a.m.).
Stretch	PyTorch residual correction model (physics)	Corrects a known HAND limitation; optional, system must run without it.
Stretch	PyTorch historical-decision scorer (tactics)	Adds a quantitative signal alongside the critic; defaults to nearest-neighbor similarity if training data proves too sparse.
Post-MVP	Chatbot tab	Adds conversational convenience only — training mode already delivers the core critique experience without it.

Assumptions

- Government hydrology data (HAND rasters, NWM streamflow, USGS gauge feeds) is sufficient for a credible demo without a trained inundation model as the primary source.
- The AAR corpus (a handful of Texas flash-flood after-action reports) is large enough for retrieval-grounded critique, but too small to reliably train a deep predictive model — any trained component is a bounded, clearly-labeled stretch goal, not a core claim.
- Physics and rendering can be fully decoupled from the LLM layer via two deliberate bypass connections (raster tiles to the map, velocity field to the search-area tool), keeping raw numerical work out of the LLM's hands entirely.
- The chatbot tab adds no new capability beyond what training mode already delivers, and is deprioritized to post-MVP accordingly.
- The system is positioned as training/pre-deployment support, not live operational guidance — this is what keeps the scope buildable in a hackathon timeframe and defensible to a technically-informed judge.

Derived from the source architecture document (flood digital twin, dnhacks September 2026) and the accompanying PRD's rendering/physics and LLM/RAG layer specifications.